

amds

Atomic & Molecular Data Selector

Version 1.0

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1. Introduction

1.1. About amsds

This code is a part of the Interactive Graphical Interface package which facilitates data preparation for two-dimensional edge plasma models like B2-Eirene.

amds allows the user to look through the files containing the reaction data, to plot them, to select a data set for the particular model, to check the consistency of the selected data, and to produce an output file which can be used by other routines to create the input files for Eirene.

1.2. About this manual

This manual is divided into several sections.

- **Introduction** contains some common description of **amds**. It is directed to all users.
- **Using amsds** describes how to use **amds**, assuming it has been properly installed. It is directed to all users.
- **Installing amsds** describes how to install **amds**. It is directed to "power users" who need to set up this program.
- **Technical info** describes file formats and the like. It is directed to developers of related programs or database files.

1.3. Files used by amsds

1.3.1. Reaction sets [*amd]

Reaction set files, or simply **amds** files, are files containing data sets selected by the user. They are used to store data sets in a format that can be written, read and further modified by **amds**.

1.3.2. Output files [*amo]

Output files can be produced by **amds** from data sets selected by the user. They are supposed to be used by programs processing **amds** output.

1.3.3. Reaction database [amds.rpd]

The reaction database is a file describing all particles, reactions and reaction types used by **amds**. This file is necessary for the operation of **amds**. Modifications to this file can result in **amds** being unable to read older reaction set files; for this reason any modifications should be done only by an expert user and with extreme care.

1.3.4. Data files [AMJUEL, HYDHEL, etc.]

Data files are text files containing atomic and molecular data. With the exception of few "extras", they are not necessary for a normal operation of **amds**.

1.3.5. Configuration [amds.rpc]

The configuration is a special file used by **amds** as a template to create new **amds** files. It should be produced by a "power user" while setting up **amds**.

2. Using amds

2.1. Starting amds

2.1.1. Setting up the X environment

amds is an X Windows application. If you run **amds** from an X terminal different from the console of the workstation, you may need to tell **amds** the address of the X terminal. This can be done either by setting (and exporting) the environment variable **DISPLAY** to the display address, or by specifying the address on the command line in the form

```
amds ... -display Address
```

Also, you may need to include the computer used to run **amds** into the list of hosts allowed to use your X terminal. See the X Windows documentation for details on display addresses and host access lists.

2.1.2. Starting with a new file

In order to start **amds** with a new file, run the executable file without specifying any parameters. A window with a new file will be opened.

```
amds
```

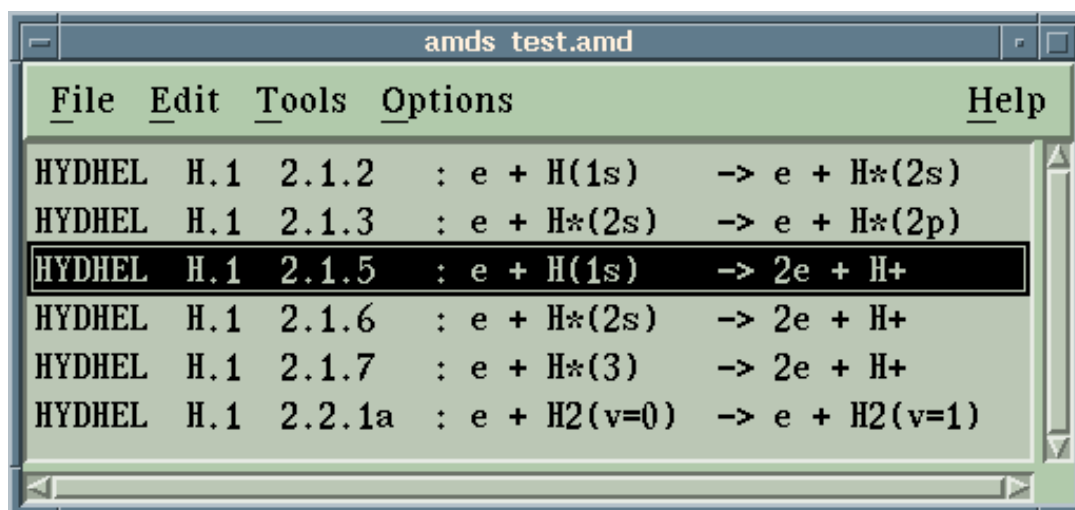
2.1.3. Starting with an existing file

In order to load existing file(s), specify filename(s) on the command line. For each filename specified, a window containing that file will be opened.

```
amds Filename ...
```

2.1.4. The main window

After you start **amds**, the **main window** will appear.



The main window consists of a menu bar, which is used to carry out different commands, and the reaction list, which contains reactions selected for the data set. The reaction list is normally empty when you start amds with a new file.

2.2. Working with files

2.2.1. Creating a new file

In order to create a new file, issue the **File|New** command from any main window. A new main window containing a new file will appear.

2.2.2. Opening a file

In order to open an existing file, issue the **File|Open** menu command from any main window. A file selection dialog box will appear. Select the desired file and press **Open** to open a new main window with this file loaded. Press **Cancel** to cancel the operation.

2.2.3. Saving a file

In order to save a file in a main window, issue the **File|Save** menu command. A file selection dialog box will appear. Select the desired filename and press **Save** to save the file. Press **Cancel** to cancel the operation.

2.2.4. Producing an output file

An output file can be produced only from a file that has a filename assigned to it. The filename is assigned either when a file is loaded from disk, or when it is saved to disk. The name of the output file is producing by changing the extension of the reactions set file to **.amo**.

In order to produce an output file, issue the **File|Output...** menu command from a main window. A question box will appear asking you for a confirmation. Press **OK** to produce an output file. Press **Cancel** to cancel the operation.

If a **missing filename** error box appears instead, this means amds cannot determine the filename because you have neither saved the file to nor loaded

it from disk. In this case, save the set of reactions under some name and issue the **File|Output..** menu command again.

2.2.5. Closing a window

In order to close a main window, either issue the **File|Close** menu command, or use the method provided by the window manager. In **mwm**, simply double-click the window system menu. **amds** will terminate when all main windows are closed.

If the file in the window has been changed since the last save, the **Unsaved work** dialog box will appear prompting you either to save the file or to discard it. Press **Save** to save the file. Press **Discard** to close the window without saving changes. Press **Cancel** to cancel the operation.

2.2.6. Exiting the program

In order to exit the program, you need to close all main windows. The **File|Exit** menu command issued from a main window can be used to close all windows. Similar to **File|Close**, it pops up **Unsaved work** dialog boxes in windows that have unsaved files.

Note: iconified windows with unsaved work are not automatically de-iconified by this command, and their warning boxes will not be visible for this reason. If you have iconified any **amds** windows, make sure all of them disappear. If some do not, de-iconify them and answer questions whether you want to save their contents or not.

2.3. Choosing reactions

2.3.1. Basics

In **amds**, so-called **subsets** are used to choose reactions. Subsets are temporary lists of reactions used for searching etc. After that, some or all reactions from a subset can be 'copied' into the main window. Besides choosing reactions to be put the main window, subsets are very useful for simply browsing the database of reactions.

Subsets are displayed in subset windows. Subset windows behave similar to main windows in many aspects. However, there are also differences.

Each subset window is **associated** with one main window. In most window managers, this causes the subset window to be displayed always on top of the main window. Also, subset windows cannot be iconified separately, but disappear together with the associated main window when it is iconified.

A main window can have more than one associated subset window.

Subset windows have a menu slightly different from the menu of a main window. Instead of the **File** menu, subset windows have a **Subset** menu, which has much less functionality. Subsets cannot be saved to or loaded from disk files. However, they can be transferred to files indirectly, by cutting and pasting subset contents into a main window and then saving the main window contents.

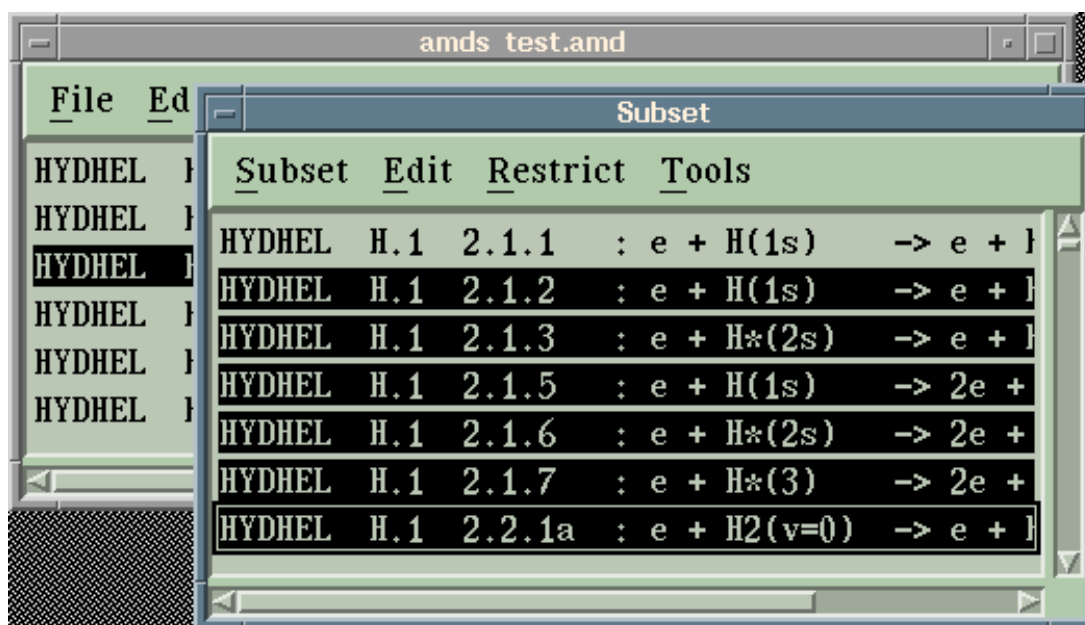
Subset windows have a **Restrict** menu. It allows restricting the list of reactions to reactions having certain properties.

Some menu commands are not available in subset windows. In this manual, it is normally indicated whether a menu command should be issued from a main or a subset window. If it is not indicated, the command performs in exactly the same way in both kinds of windows.

When choosing reaction(s) to be put into the main window, one normally opens a subset window, uses the **Restrict** menu to restrict the subset to only a few reactions, and then chooses desired reactions and 'activates' them in the main window.

2.3.2. Creating and closing subset windows

In order to create a subset window containing all reactions in the database, issue the **File|New subset** menu command from a main window. A new subset window, associated with that main window, will be opened.



In order to create a subset window containing the reactions currently contained in the main window, issue the **File|Clone subset** menu command from a main window. A new subset window, associated with that main window, will be opened.

In order to create a subset window containing the reactions currently contained in another subset window, issue the **Subset|Clone subset** menu command in the subset window. A new subset window, associated with the same main window, will be opened.

Subset windows have **Undo/Redo** commands in the menu, similar to main windows. These commands in subset window affect only changes made to the subset, and not changes made in the main window. Similarly, undo/redo commands in the main window do not affect any subset windows associated with it.

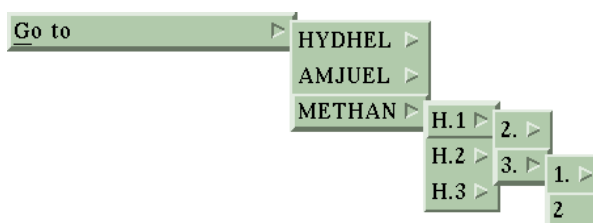
In order to close a subset window, issue the **Subset|Close** menu command, or use the method provided by the window manager.

Closing a main window automatically closes all associated subset window.

2.3.3. Navigating through the list of reactions

You can scroll through the list of reactions using the scrollbar or keyboard commands.

The **Edit|Goto** system of submenus allows you jumping directly to a reaction identified by its database, section and number. In the submenu, first select the database name, then the section number, then the reaction number by picking its page number. Reactions not present in the list are



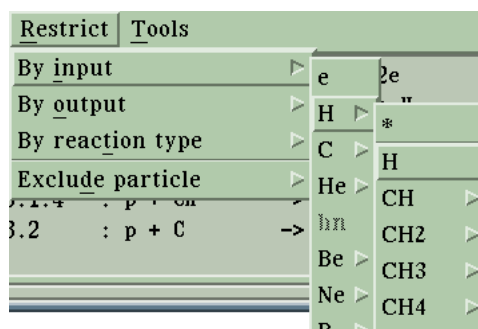
2.3.4. Restricting the list of reactions

In subset windows, it is possible to restrict the list of reactions to reactions containing a specific input or output particle or a class of particles, or to reactions of a specific type.

In order to restrict the list of reactions to reactions containing a specific input particle, open the **Restrict|By input** menu tree. This menu tree consists of regular menu entries, representing regular particles, and submenus, representing particle classes. In order to select one particle, find it in the menu tree and select it's menu entry. In order to select a whole class or particles, find it's submenu and select the * menu entry in that submenu. Note that this will also select indirectly subclassed particles.

For example, if **He+** is subclass of **He**, and **He+(1s)** a subclass of **He+**, selecting **He|*** will also include **He+(1s)**.

In order to restrict the list according to input particles, choose the particle or the class of particles in the **Restrict|By input** menu tree. Only reactions having that input particle or an input particle of that class will be left in the list.



In order to restrict the list according to output particles, use the **Restrict|By output** menu tree in the similar way.

In order to restrict the list to reactions **not** containing a particle or a particle of some class, use the **Restrict/Exclude particle** menu tree in the similar way.

In order to restrict the list to reactions of a certain type, choose the type in the **Restrict|By type** menu.

Menu entries for choices that would cause all reactions to be removed are greyed out and cannot be selected.

It is very convenient to use the **Undo** command to reverse the effect of a restrict operation. For example, one could restrict the list to a few reactions, use the Activate command to put some of them into the main window, and use Undo to return to the full list. Because the **Activate** command modifies the **main** window, its effect is left intact after **Undo**. All this eliminates the need for frequently opening new subset windows.

2.3.5. Activating and deactivating reactions

From a subset window, you can add or remove reactions to/from the associated main window.

In order to add reaction(s) to the main window, select them in the reaction list of the subset window and issue the **Edit|Activate** menu command in the subset window.

In order to remove reaction(s) from the main window, select them in the reaction list of the subset window and issue the **Edit|Deactivate** menu command in the subset window. Often it is easier to use the **Cut** command in the main window instead, as described below.

Edit	Restrict	Tools
Undo	Ctrl+Z	
Redo	Ctrl+Shift+Z	
Redo all		
Activate	Ctrl+A	
Deactivate	Ctrl+D	
Cut		
Copy		
Crop		
Paste		
Go to		

Note that **Activate/Deactivate** menu commands modify the contents of the main window, and not the contents of the subset window. In order to reverse their effect, **Undo/Redo** commands must be issued in the main window!

2.3.6. Cut & Paste

amds internally maintains a special subset called **clipboard**. It is not visible in any window, and it is not associated with any window. Using some commands from the Edit menu, it is possible to modify the content of the clipboard, or insert reactions currently on the clipboard into a reaction list. These menu command work in the same way in main and subset windows.

The **Edit|Cut** menu command clears the clipboard and copies the reactions currently selected in the reaction list to the clipboard. Then, it removes these reactions from the list. Besides its use for clipboard operations, this command is convenient for simply deleting unwanted reactions from the list.

The **Edit|Copy** menu command clears the contents of the clipboard and copies the reactions currently selected in the reaction list to the clipboard. It does not modify the reaction list in the window.

The **Edit|Crop** menu command clears the contents of the clipboard and copies the reactions currently **not** selected in the reaction list to the clipboard. Then, it removes these reactions from the list. Besides its use for clipboard operations, this command is convenient for leaving only selected reactions on the list.

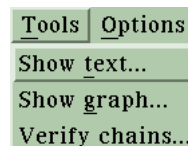
The **Edit|Paste** menu command adds all reactions currently on the clipboard to the reaction list in the window. It does not modify the clipboard contents.

Undo/Redo commands in the window only affect the modifications made to the reaction list. There is no possibility to reverse the effect of modifications to the clipboard.

2.4. Using tools

2.4.1. Displaying data files

amds can display the location in a data file (from text files **HYDHEL**, **AMJUEL**, etc.) corresponding to a reaction. This is possible both in main and subset windows.



In order to do this, select the desired reaction in the list and issue the **Tools|Show text** menu command. Note that this command will have no effect if not exactly one reaction in the list is selected.

Alternatively, you can double-click a reaction in the list to display its text.

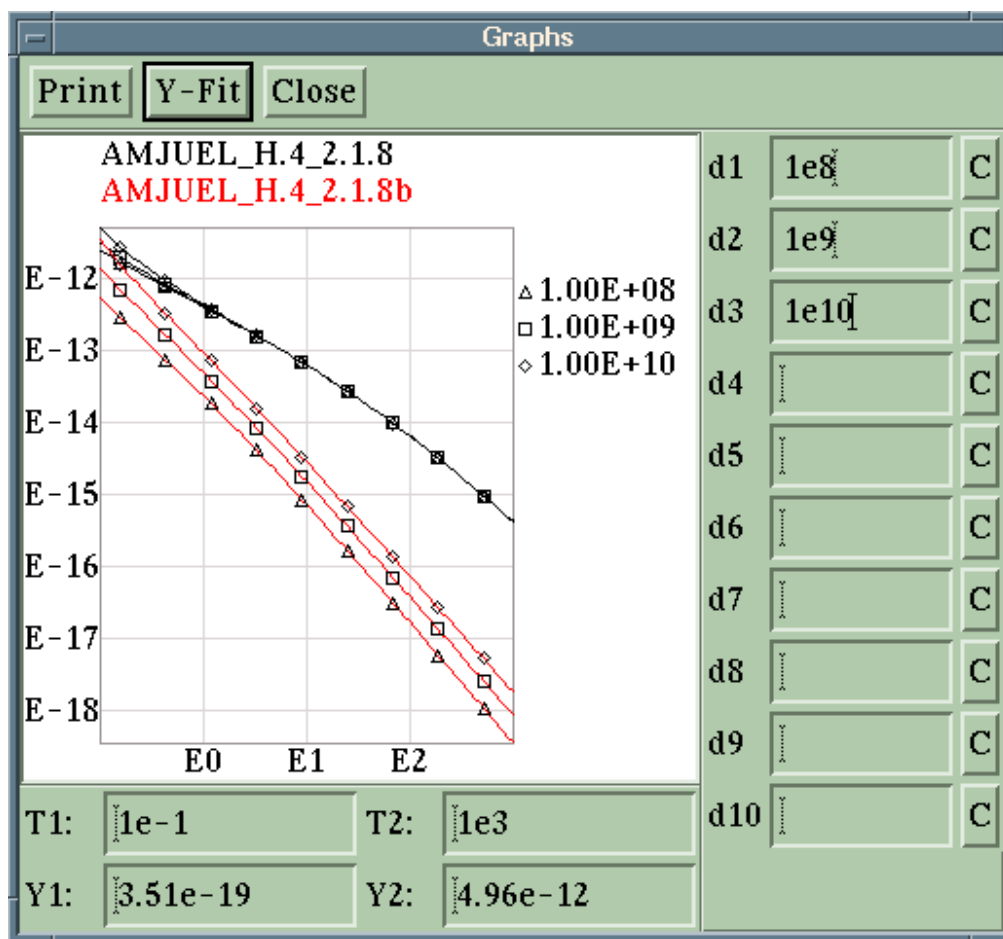
A window containing the corresponding data file will appear, and the base text will be scrolled so that the description of the selected reaction will be visible in the window.

In order to close that window, either use the **Window|Close** menu command, or use the window manager. You can continue working with **amds** with this window left open.

2.4.2. Displaying graphs of reaction data

amds can be used to display and print graphs of reaction data. An external program is used to calculate the data; it must be set up properly before you can display graphs. See notes on installing **amds** for details.

In order to display or print graphs, issue the **Tools|Show graph...** menu command. The **Graphs** dialog box will appear. It will be associated with the window where the menu command was issued.



In order to display a graph of reaction data, select the desired reactions in the window associated with the **Graphs** dialog box and specify desired densities in **d1..d10** text edit fields. Leave unused edit fields empty.

amds will run an external program in background, which will calculate reaction data. After the program finishes, **amds** displays a graphical representation of it's output. Reaction data are recalculated each time some values in the **Graphs** dialog box are changed, or different reactions selected in the list. There is no need to press any button to update the graph.

The physical meaning of reaction data is unimportant to **amds** and is determined by the program used to calculate the data. In **amds**, the values **T1..T2** are referred to as 'temperature', and the values **d1..d10** as 'densities'.

Graphs for different reactions are represented using different colors. Graphs for different densities are represented using different centered symbols. A logarithmic coordinate system is used for both directions, and the leading '1' is omitted from coordinates displayed in the graph window.

In order to change the temperature range, modify the values in **T1** and **T2** text edit boxes. This will cause **amds** to re-calculate reaction data.

In order to change the vertical range, modify the values in **Y1** and **Y2** text edit fields. Also, you can press the **Y-Fit** button in order to make all graphs fully visible. This button will put appropriate values in **Y1** and **Y2** fields. Changing the vertical range does not cause any re-calculations.

In order to add a graph with a new density, type the density in some empty **d...** field. New reaction data will be calculated and the new graph appear.

In order to change some density, modify the density in some **d...** field. New reaction data will be calculated and the new graph appear.

In order to remove some density, clear the corresponding **d...** field. The easiest way to do this is to press the **C** button located next to the field.

In order to display graphs for other reactions, select desired reactions in the window associated with the dialog box. Graphs will be updated automatically.

You can print the graphs currently visible in the **Graphs** dialog box on a PostScript printer, or generate a PostScript file. Pressing the **Print** button will open the **Print** dialog box.

Use the **To file/Via command** option menu to select the way PostScript output is made.

Select **To file** to generate an Encapsulated PostScript file. Type the filename in the text edit field next to the option menu.

Select **Via command** to run a Unix command with PostScript output as it's standard input. This can be useful for sending PostScript directly to the printer. Type the name and arguments of the command in the text field next to the option menu.

Use text fields to select page dimensions and the margin. All values should be specified in inches. The margin is subtracted from all four page sides. The default setting is for the **A4** paper size.

Press **Print** to generate the PostScript output. Press **Cancel** to close the dialog box without printing.

In order to close the **Graphs** dialog box, press the **Close** button. You can continue working with **amds** without closing the **Graphs** dialog box, but additional calculations for updating it would lead to slightly slower response from **amds**.

If after a few seconds, no graphs appear in the **Graphs** dialog box despite selecting reactions in the list and specifying densities, the problem could be related to the program used to calculate reaction data. **amds** normally prints error messages when it cannot execute that program for some reason. Check the terminal used to start **amds** for any error messages of this kind.

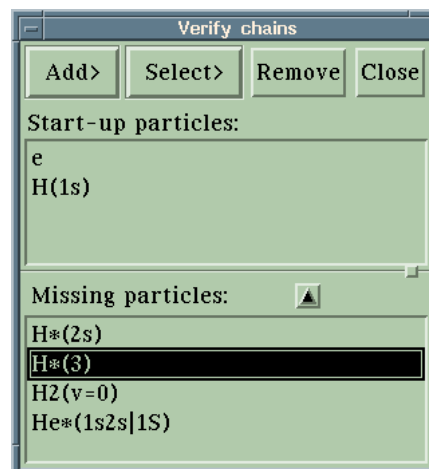
2.4.3. Verifying reaction chains

amds can do a very simple verification that each reaction in a reaction set can happen. You can specify a list of so-called **start-up particles**, which are present at the beginning. **amds** verifies that input particles needed for each reaction are either start-up particles, or can be produced by reactions currently in the list from other particles.

In order to do the verification, issue the **Tools|Verify chains...** menu command from a main window. The **Verify chains** dialog box will appear.

This dialog box has two list boxes. The start-up particles list box lists all start-up particles you selected. You can modify its contents using control elements of the dialog box.

The missing particles list box lists all particles that are needed for some reactions to happen, but cannot be produced by reactions currently in the reaction list. amds automatically updates this list each time the list of start-up particles or the set of reactions is changed. The goal of the verification is to ensure there are no missing particles.



In order to add a particle to the list of start-up particles, select desired particle(s) in the missing particles list box and press the **^\$** button.

Alternatively, you can use the **Add>** cascade button. A menu tree for selecting a particle, similar to submenus in the **Restrict** menu of a subset window, will appear. Select a particle or a class of particles in the menu in order to add them to the list of start-up particles.

In order to remove particle(s) from the list of start-up particles, select them in this list and press the **Remove** button.

Besides selecting list items with keyboard or mouse, you can use the **Select>** cascade button in order to select a particle or a class of particles in the list of start-up particles. When you press this button, a menu tree similar to that produced by the **Add>** button will appear.

In order to close the dialog box, press the **Close** button or use the window manager.

The list of start-up particles is considered to be a part of the file. It is saved to disk and loaded from disk together with the reactions set. The changes made to this list can be reversed/repeated using **Undo/Redo** commands in the corresponding main window. The contents of this list are preserved when the dialog box is closed.

2.5. Working with variables

2.5.1. Caution

Variables are not a "finished" feature yet, both in the program and in the manual.

2.5.2. Defining variables

It is possible to define variables related to reactions, or to input particles in reactions. In order to define variables, issue the **Options|Setup...** menu

command from a main window and select **Variable Definitions** in the dialog box that will appear. Use dialog boxes to add, remove or change definitions of variables.

2.5.3. Changing reaction variables

In order to change reaction variables, issue the **Options|Setup...** menu command from a main window and select **Reaction Variables**. Use the dialog box that will appear to set variables related to reactions selected in the reaction list.

2.5.4. Changing particle variables

In order to change particle variables, issue the **Options|Setup...** menu command from a main window and select **Particle Variables**. Use the dialog box that will appear to set variables related to input particles in reactions currently selected in the reaction list.

3. Installing amds

3.1. Compiling

amds has been successfully compiled and tested on IBM RISC, SGI and DEC Alpha-based workstations. Because no machine-dependent code is used, it is likely that it will compile on other platforms as well. However, it is possible that some unknown bugs, while producing no effects on these 3 platforms, will prevent **amds** from compiling or working properly.

The **makefile** included in the source code should do all the compilation and linking. Following files are needed to build the program:

res2fbr.c	Auxiliary program used to build fallback resources from amds.res .
x_~fbr.c	Fallback resources, automatically built by res2fbr .
Other *.c, *.h	Source files used to build amds .
amds.res	Resources for X Windows elements and customization.
makefile	Makefile to build amds .

3.2. Setting up the reaction database

The reaction database file is absolutely necessary to run **amds**, and the program will refuse to start without this file. A proper database file needs to be written before amds could have any practical use. The example database included in the **amds** archive was produced by an automatic macro, is incomplete and **not** usable for any scientific purposes. The format of the database file is described in '**Technical info**'.

By default, the name of the reaction database file is produced by appending '.rpd' to the name of amds executable file. It is possible to specify another file location either by inserting a line

AMDS.databaseFile: *Directory*

into the **.Xdefaults** file, or by setting the **AMDSDATABASEFILE** environment variable to the proper value. The setting in **AMDSDATABASEFILE** overrides settings in **.Xdefaults**.

If **amds** is unable to locate the database file, the program will refuse to start.

3.3. Setting up configuration

When **amds** starts with a new file or a **File|New** command is used, **amds** does not really create an empty file, but loads a special **amds** file called **configuration**. The name of the configuration file is produced by appending **'.rpc'** to the name of the **amds** executable file.

The configuration file should contain settings that should be default for all new files. For example, it can contain some start-up particles that are always present, or definitions of some variables.

Currently, changes in the configuration file only affect new files created using **amds**. The configuration does not affect files that are loaded from disk. This will change in future versions of **amds**.

If **amds** does not find the configuration file, it aborts with an error message. It is possible to instruct **amds** not to load the configuration file using the following switch:

amds -nocfg

This switch should only be used the first time **amds** is started. Then, a configuration file should be created and used.

In order to create the configuration file, use the **File|Save** menu command in a way similar to saving normal files. Specify the name of the configuration file in the dialog box, for example **amds.rpc**.

3.4. Setting up data files

All data files should be in one directory. Their names must be identical with names of databases used in the reaction database, including the case of alphabetic characters.

By default, **amds** tries to load these files from the directory containing the **amds** executable file. It is possible to specify another directory either by inserting a line

AMDS.databaseDir: *Directory*

into the **.Xdefaults** file, or by setting the **AMDSDATABASEDIR** environment variable to the proper value. The setting in **AMDSDATABASEDIR** overrides settings in **.Xdefaults**.

If **amds** is unable to locate database text files, the program will not crash, but the **Show text** menu command will produce no effect.

3.5. Setting up reaction data calculation

amds relies on an external program for calculating reaction data used to display graphs. You need to specify the name of this program and its working directory. See '**Technical info**' for detailed information on calling conventions for this program.

The name of the program can either be specified in the **.Xdefaults** file:

AMDS.graphCmd: *ProgramName*

or in the **AMDSGRAPHCMD** environment variable. The value in the environment overrides the setting in **.Xdefaults**. If no name is specified, **./calc_atomic_data** is used.

The working directory of the program can either be specified in the **.Xdefaults** file:

AMDS.graphDir: *Directory*

or in the **AMDSGRAPHDIR** environment variable. The value in the environment overrides the setting in **.Xdefaults**. If no directory is specified, **.'** is used.

4. Technical info

4.1. Database file format

Following objects can be defined in the database:

Particles

Properties:

- Name
- Superclasses

Particles should be organized into a hierarchical structure in order to simplify selecting particles and reactions from the menu. This structure does not need to have any physical meaning. Each particle can have one or more so-called superclasses, which are used to create cascading submenus. The top-level superclass is called 'Particle'.

Particle classes

Properties:

- Name
- Superclasses

Particle classes are used to organize particles into a hierarchical structure. They do not need to have any physical meaning.

Reaction types

Properties

- Name

Reaction types are used to identify kinds of reactions. Each reaction can belong to any number of reaction types. Each reaction has a so-called primary type that is passed to the program calculating reaction data and may have other special meanings in the future.

Reactions

Properties:

- Database name
- Section name
- Reaction number
- Input particles - ordered
- Output particles - ordered
- Reaction type(s)

Reactions are identified by the combination of database and section names and the reaction number.

Database files are pure ASCII text files. Each line must start with a keyword which specifies the kind of the object defined in that line. The keyword should be followed by keyword-specific parameters, which should be separated by spaces except where noted. Following keywords are currently supported:

Particle *Name* [: *Superclass* [*Superclass*] ...]

Declares a particle. If no superclasses specified, it becomes a subclass of 'Particle'.

Name Particle name.

Superclass Superclass name. This can be another particle, or a particle class name.

ParticleClass *Name* [: *Superclass* [*Superclass*] ...]

Declares an abstract particle class. If no superclasses specified, it becomes a subclass of 'Particle'.

Name Particle class name.

Superclass Superclass name. This can be another particle class or a particle name.

ReactionType *Name*

Declares a reaction type.

Name The name of the reaction type.

Reaction *Database* *Section* *Number* : [Count]*InputParticle* ... : [Count]*OutputParticle* ... : *Type* ...

Declares a reaction.

Database The name of the database the reaction comes from.

<i>Section</i>	The name of the section of the database.
<i>Number</i>	The number of the reaction in it's section.
<i>Count</i>	Used to specify how many particles of the same kind take part in the reaction. By default 1. Note that no space between <i>Count</i> and <i>...Particle</i> is allowed.
<i>InputParticle</i>	The name of an input particle. Optionally can be prefixed by <i>Count</i> . Multiple input particles can be specified.
<i>OutputParticle</i>	The name of an output particle. Optionally can be prefixed by <i>Count</i> . Multiple output particles can be specified.
<i>Type</i>	The name of a reaction type. Multiple reaction types can be specified. The first type specified becomes the primary type for the reaction.
<i># Comment</i> <i>; Comment</i>	

Inserts a comment. Comments are ignored by amds. Also, empty lines are ignored.

Comment Any text.

Other keywords are simply ignored without any error message. They should not be used in order to prevent problems with future versions of amds.

4.2. Output file format

Output files are pure ASCII files. Some lines in the output file start with special keywords. These keywords may also indicate the meaning of lines following the line with a keyword. Following keywords are available:

Comment

A comment. Comments should be ignored by the program reading the output file.

Comment Any text.

fileversion *Version*

Specifies the version of the output file format. amds v1.0 produces version 1.

Version An integer number specifying the version of the output file format.

reactions *Count*
Database Section Number
...

Specifies the reactions selected. The **reactions** keyword is followed by *Count* lines, each line identifying one reaction.

<i>Count</i>	The number of reactions to follow.
<i>Database</i>	Database name, as specified in the reaction database.
<i>Section</i>	Section name, as specified in the reaction database.
<i>Number</i>	Reaction number, as specified in the reaction database.
<i>VarName</i>	
<i>Value</i>	
...	

Specifies values of a variable. Only variables with the setting **No Export** disabled are written into the output file. The *VarName* keyword is followed by a number of lines, each line containing a value for one instance of the variable.

For variables related to reactions, the number of values is equal to the number of selected reactions. The order of values corresponds to the order of reactions in the output file.

For variables related to particles in reactions, the number of values for each reaction is equal to the number of different input particles for that reaction. Using variables related to particles in reactions is not recommended.

<i>VarName</i>	The name of the variable, as specified in the Create Variable dialog box.
<i>Value</i>	A value of the variable. Each value is output on a separate line.

finish

Specifies the end of the output file. Additional text may follow this statement; this text does not contain any valuable information.

4.3. Calling conventions for reaction data calculation

The program for calculating reaction data is started with the following command-line parameters:

Programe Database Section Reaction Type T1 T2 D1 [D2] ...

<i>Programe</i>	Program name
<i>Database</i>	Database name, as in the reaction database.
<i>Section</i>	Section name, as in the reaction database.
<i>Reaction</i>	Reaction name, as in the reaction database.
<i>Type</i>	Reaction type, as the primary type in the reaction database.
<i>T1</i>	Start temperature.
<i>T2</i>	End temperature.

D . . . Densities. The program must be able to accept at least 10 densities as parameters.

The program must be reliable and produce some results for **any** possible values of it's arguments. In particular, it should terminate in a finite and short amount of time even in case of incorrect arguments. The program should not display any error messages if it encounters incorrect numeric arguments, because incorrect arguments **are** passed to it when the user has not finished typing values in text fields.

The program should either create no temporary files, or catch the SIGTERM signal and destroy these files in that case.

The program must write calculated reaction data into the standard output. Reaction data must be prefixed by a line

start of data

All text before this line is ignored by **amds**. This line must be immediately followed by a number of lines containing calculated data in the form

T V1 [V2] . . .

T Temperature.

V1 . . . Cross-sections corresponding to the various densities. The number and order of these values on each line must be identical to the number and order of densities passed on the command line.

Reaction data must be immediately followed by a line

end of data

All text following this line is ignored by **amds**.